

NEWCASTLE WILLIAMTOWN, Australia

WMO: 947760

Lat: 32.7933S

Lon: 151.8358E

Elev: 26

StdP: 14.68

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 39.0	(c) 41.0	(d) 25.1	(e) 19.2	(f) 64.7	(g) 28.7	(h) 22.7	(i) 60.0	(j) 31.4	(k) 60.3	(l) 28.4	(m) 58.2	(n) 4.3	(o) 300	(p) 0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 17.9	(c) 95.4	(d) 69.5	(e) 90.8	(f) 69.2	(g) 86.8	(h) 69.0	(i) 74.8	(j) 85.3	(k) 73.5	(l) 82.7	(m) 72.4	(n) 80.7	(o) 14.6	(p) 300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.1	118.9	77.5	71.1	114.9	76.6	69.9	110.2	75.5	38.3	85.3	37.1	82.9	36.1	80.7	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 26.3	(b) 22.9	(c) 20.5	(e) 33.7	(f) 104.8	(g) 1.9	(h) 3.7	(i) 32.3	(j) 107.5	(k) 31.2	(l) 109.6	(m) 30.1	(n) 111.7	(o) 28.7	(p) 114.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	74.2	73.4	70.4	64.9	59.5	55.4	53.8	55.4	60.2	64.4	68.2	71.5
	DBStd	8.39	5.22	4.62	4.28	3.81	3.76	3.34	3.17	3.96	4.82	5.31	5.83	5.49
	HDD50	8	0	0	0	0	0	2	4	2	0	0	0	0
	HDD65	1440	0	0	5	46	177	289	348	300	161	78	29	7
	CDD50	5194	751	656	632	446	293	163	121	168	306	445	546	667
	CDD65	1150	287	236	172	41	5	0	0	1	16	58	125	209
	CDH74	8476	2207	1586	887	251	31	0	2	33	259	626	1020	1574
	CDH80	3090	909	535	226	47	1	0	0	4	74	235	427	632
Wind	WSAvg	9.4	10.2	9.6	8.9	7.8	8.6	9.4	9.7	9.9	9.7	9.6	9.9	10.1
Precipitation	PrecAvg	44.8	3.6	4.4	5.0	4.6	4.4	5.1	2.6	2.6	2.4	2.9	3.3	3.2
	PrecMax	70.6	12.8	23.6	15.7	14.3	15.8	15.5	7.5	8.3	7.0	9.4	9.5	9.4
	PrecMin	21.3	0.1	0.2	0.1	0.2	0.2	0.6	0.0	0.0	0.0	0.0	0.4	0.6
	PrecStd	10.6	2.8	3.9	3.3	3.7	3.2	3.3	1.8	2.0	1.8	2.0	1.9	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	102.5	99.0	91.6	85.8	77.8	70.5	72.3	78.7	87.7	93.1	98.2	100.2
		MCWB	70.2	72.2	70.8	65.9	61.7	59.2	56.7	59.8	60.4	62.0	67.2	68.8
	2%	DB	94.9	90.7	85.5	79.9	73.6	67.3	67.6	73.0	81.2	86.7	90.4	92.3
		MCWB	71.6	72.6	70.0	65.3	60.9	56.9	54.0	56.0	59.8	62.6	67.2	69.3
	5%	DB	89.3	86.1	81.5	76.5	70.5	65.2	64.6	68.9	76.0	80.8	84.3	86.7
		MCWB	71.3	71.2	69.3	64.7	60.1	56.2	53.4	54.5	58.6	63.0	66.3	69.4
	10%	DB	84.3	82.1	78.5	73.6	68.2	63.1	62.4	65.8	71.4	75.7	79.1	82.1
		MCWB	70.9	70.6	68.6	63.9	59.3	55.6	52.9	53.9	57.9	62.0	66.1	68.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.5	77.0	74.8	70.1	66.7	62.3	61.3	62.3	65.5	69.2	73.4	75.0
		MCDB	90.0	88.2	82.6	76.6	71.1	65.9	66.3	72.6	75.0	79.4	87.4	86.7
	2%	WB	74.7	74.8	72.7	68.3	64.6	60.5	58.4	59.5	63.6	67.3	71.0	72.8
		MCDB	86.1	83.9	79.7	74.4	69.5	64.4	62.6	66.9	72.4	76.1	82.0	83.5
	5%	WB	73.5	73.4	71.4	67.0	62.7	59.1	56.4	57.7	62.0	65.7	69.2	71.5
		MCDB	83.1	81.1	78.0	73.3	68.3	62.8	61.0	64.7	70.1	74.1	78.3	80.7
	10%	WB	72.4	72.0	70.2	65.6	60.9	57.6	54.8	56.0	60.3	64.2	67.6	70.1
		MCDB	80.7	79.1	76.0	71.9	66.2	61.3	59.7	62.9	68.1	72.1	75.6	78.3
Mean Daily Temperature Range	5% DB	MDBR	17.9	16.6	16.6	18.3	18.2	16.8	18.8	21.3	21.8	21.1	18.8	18.4
		MCDBR	28.8	25.0	22.6	23.5	21.8	19.3	23.1	28.0	31.1	32.0	29.8	28.2
		MCWBR	9.8	9.2	9.6	11.2	11.4	11.1	12.6	13.6	13.2	12.8	10.9	10.0
	5% WB	MCDBR	21.4	19.7	17.7	19.2	17.7	15.6	17.9	22.4	24.1	23.2	22.7	22.2
		MCWBR	8.3	8.3	8.4	10.4	10.5	10.3	11.6	12.8	12.7	11.3	9.7	9.1
Clear-Sky Solar Irradiance	taub	0.397	0.387	0.356	0.334	0.316	0.311	0.301	0.309	0.345	0.364	0.387	0.385	
	taud	2.447	2.496	2.578	2.609	2.609	2.618	2.630	2.596	2.486	2.466	2.445	2.470	
	Ebn at Noon	299	295	294	284	273	267	276	288	292	299	300	304	
	Edh at Noon	38	35	30	26	23	22	23	26	33	36	38	38	
All-Sky Solar Radiation	RadAvg	2097	1846	1553	1244	978	782	912	1206	1585	1825	1963	2110	
	RadStd	134	153	134	97	75	64	74	75	85	148	182	165	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.76	+0.83	N/A	N/A	+0.70	N/A	N/A	-135	+331
(47) Regional (29 neighbors)		+0.77	N/A	N/A	+1.39	N/A	N/A	-2	-153	+342

Nomenclature: See separate page